

Problems on phase diagrams

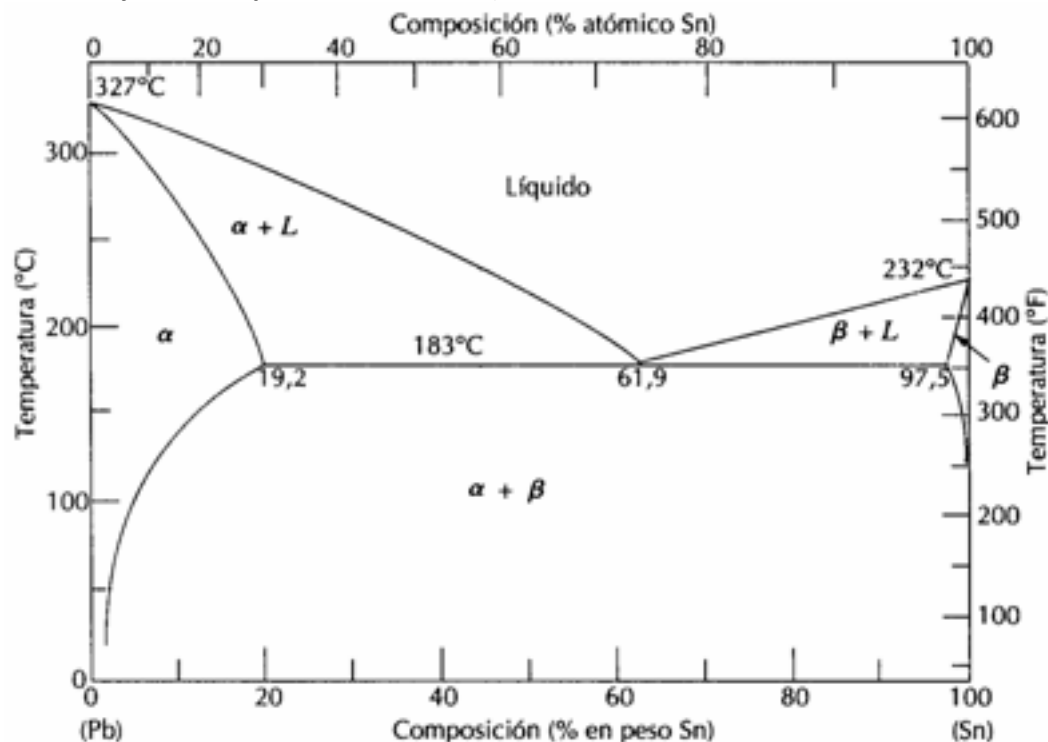
1. In the phase diagram of two hypothetical metals, A and B, there is a phase α rich in A and a phase β rich in B. Given the mass fraction of two alloys for a specific temperature (table), determine the composition of the phase limits (or solid solubility limits) of the α and β phases at such temperature.

Solution: $C_\alpha = 11.7\%$ $C_\beta = 95\%$

Alloy composition	Fraction of α phase	Fraction of β phase
30% B	0.78	0.22
65% B	0.36	0.64

2. From 2.8 kg of a Sn-Pb alloy, is it possible to have 2.21 kg of primary β and 2.53 kg of β total at $(183-\Delta T)^\circ\text{C}$? Justify your answer.

Solution: Yes, for an alloy with composition $C_0 = 90\% \text{ B}$.



3. Using the Pb-Sn phase diagram for an alloy composition of Pb-80%Sn, indicate at the temperatures of $T_1 = 250^\circ\text{C}$, $T_2 = 183 + \Delta T$, $T_3 = 183 - \Delta T$, $T_4 = 100^\circ\text{C}$

- The phases present, chemical composition of the phases and mass fraction.
 - The micro constituents, chemical composition and mass fractions.
- Note the agreement that exists between the data obtained in (a) and (b).

Solutions: a) T_1 : $C_L = 80\% \text{ Sn}$, $w_L = 1$, T_2 : $C_L = 61.9\% \text{ Sn}$, $C_\beta = 97.5\% \text{ Sn}$, $w_L = 0.49$, T_3 : $C_\alpha = 19.2\% \text{ Sn}$, $C_\beta = 97.5\% \text{ Sn}$, $w_\beta = 0.22$; T_4 : $C_\alpha = 5\% \text{ Sn}$, $C_\beta = 100\% \text{ Sn}$, $w_\beta = 0.21$; b) $C_\beta = 97.5\% \text{ Sn}$, $C_{\beta+\beta'} = 61.9\% \text{ Sn}$, $w_{\beta+\beta'} = 0.49$, $w_\beta = 0.51$

4. Using the Pb-Sn phase diagram for an alloy composition of Pb-30%Sn, indicate at the temperatures of $T_1 = 250^\circ\text{C}$, $T_2 = 183 + \Delta T$, $T_3 = 183 - \Delta T$, $T_4 = 100^\circ\text{C}$

- The phases present, chemical composition of the phases and mass fraction.
- The micro constituents, chemical composition and mass fractions.

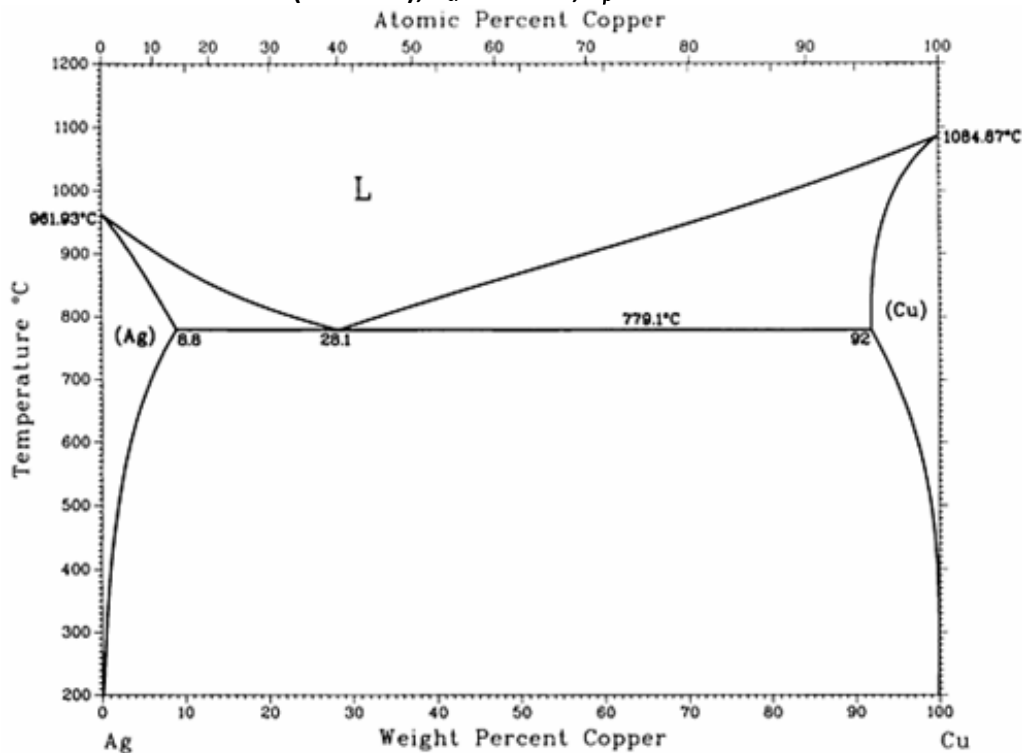
Note the agreement that exists between the data obtained in (a) and (b).

Solutions: a) T_1 : $C_L = 36\% \text{ Sn}$, $C_\beta = 12\% \text{ Sn}$, $w_L = 0.75$, T_2 : $C_L = 61.9\% \text{ Sn}$, $C_\beta = 19.2\% \text{ Sn}$, $w_L = 0.25$, T_3 : $C_\beta = 19.2\% \text{ Sn}$, $C_\alpha = 97.5\% \text{ Sn}$, $w_{\beta\alpha} = 0.86$; T_4 : $C_\beta = 5\% \text{ Sn}$, $C_\alpha = 100\% \text{ Sn}$, $w_{\beta\alpha} = 0.74$; b) T_3 : $C_\beta = 19.2\% \text{ Sn}$, $C_{\beta\alpha} = 61.9\% \text{ Sn}$, $w_{\beta\alpha} = 0.25$, $w_\beta = 0.75$

5. Given an alloy Ag-28.1%Cu, indicate for the given temperatures, $T_1 = 900^\circ\text{C}$; $T_2 = (780 + \Delta T)^\circ\text{C}$; $T_3 = (780 - \Delta T)^\circ\text{C}$:
- Present phases, chemical composition of phases and mass fraction.
 - Microconstituents, chemical composition and mass fraction at T_3 .

Solution:

- a) T_1 : Liquid (28.1%Cu); T_2 : Liquid (28.1%Cu); T_3 : $\alpha(C_\alpha = 8.8\% \text{ Cu}) + \beta(C_\beta = 92\% \text{ Cu})$; $x_\alpha = 0.77$; $x_\beta = 0.23$
 b) Eutectic microconstituent (28.1%Cu); $C_{\alpha E} = 8.8\% \text{ Cu}$; $C_{\beta E} = 92\% \text{ Cu}$



6. Using the Cu-Ag phase diagram, for an alloy with composition $C_0 = 80\% \text{ Cu}$, for temperatures $T_1 = 900^\circ\text{C}$; $T_2 = (780 + \Delta T)^\circ\text{C}$; $T_3 = (780 - \Delta T)^\circ\text{C}$, determine:
- Present phases, chemical composition of phases and mass fraction.
 - Microconstituents, chemical composition and mass fraction at T_3 .

Solution:

- a) T_1 : $C_L = 58\% \text{ Cu}$; $C_\beta = 93\% \text{ Cu}$; $x_L = 0.37$; $x_\beta = 0.63$; T_2 : $C_L = 28.1\% \text{ Cu}$; $C_\beta = 92\% \text{ Cu}$; $x_L = 0.19$; $x_\beta = 0.81$; T_3 : $C_\alpha = 8.8\% \text{ Cu}$; $C_\beta = 92\% \text{ Cu}$; $x_\alpha = 0.14$; $x_\beta = 0.86$
 b) eutectic microconstituent ($x_E = 0.19$) and proeutectic β ($x_{\beta p} = 0.81$)

7. Using the Cu-Ag phase diagram, for an alloy with composition $C_0 = 10\% \text{ Cu}$, for the temperatures $T_1 = 900^\circ\text{C}$, $T_2 = (780 + \Delta T)^\circ\text{C}$, $T_3 = (780 - \Delta T)^\circ\text{C}$, determine:
- Present phases, chemical composition of phases and mass fraction.
 - Microconstituents, chemical composition and mass fraction at T_3 .

Solution:

- a) T_1 : Liquid (10%Cu); T_2 : $C_L = 28.1\% \text{ Cu}$; $C_\alpha = 8.8\% \text{ Cu}$; $x_\alpha = 0.94$; $x_L = 0.06$; T_3 : $C_\alpha = 8.8\% \text{ Cu}$; $C_\beta = 92\% \text{ Cu}$; $x_\alpha = 0.98$; $x_\beta = 0.02$;
 b) eutectic microconstituent ($x_E = 0.06$) and proeutectic α ($x_{\alpha p} = 0.94$)

8. Two metals (A and B) have partial solubility among each other. Metal A, with a melting point at 700°C, can dissolve 5% of B at 0°C, and its maximum solubility occurs at 400°C, being 25% B. On the other hand, B metal melts at 600°C, can dissolve 10% of A at 0°C and its maximum solubility occurs at 400°C, being 20% A. At 400°C there is an eutectic reaction when the B content is 65%.

- Draw and identify the parts of the phase diagram in equilibrium (suppose straight lines)
- For an alloy 40% B, determine the phases composition and their relative proportions at 450°C, 400°C+ΔT and 400°C-ΔT.
- For an alloy 40% B, draw its microstructure indicating the microconstituents and the mass fraction of each one. Additionally, indicate the mass fraction of each of the phases present in such microconstituents.

Solution:

b) 450°C: $C_L=55\%B$; $C_\alpha=20\%B$; $X_\alpha=0.43$; $X_L=0.57$; **400°C+ΔT:** $C_L=65\%B$; $C_\alpha=25\%Cu$; $X_\alpha=0.62$; $X_L=0.38$; **400°C-ΔT:** $C_\beta=80\%B$; $C_\alpha=25\%B$; $X_\alpha=0.73$; $X_\beta=0.27$

c) Eutectic microconstituent and proeutectic α; $X_{\alpha p}=0.62$; $X_E=0.38$; **Phases in the eutectic microconstituent:** $X(\alpha_E)=0.11$; $X(\beta_E)=0.27$.

9. Given the Al-Ni phase diagram, answer the questions below.

- What is the maximum solubility of aluminum in solid state nickel? What about nickel in aluminum?

Solution: 10%Al; 1%Ni

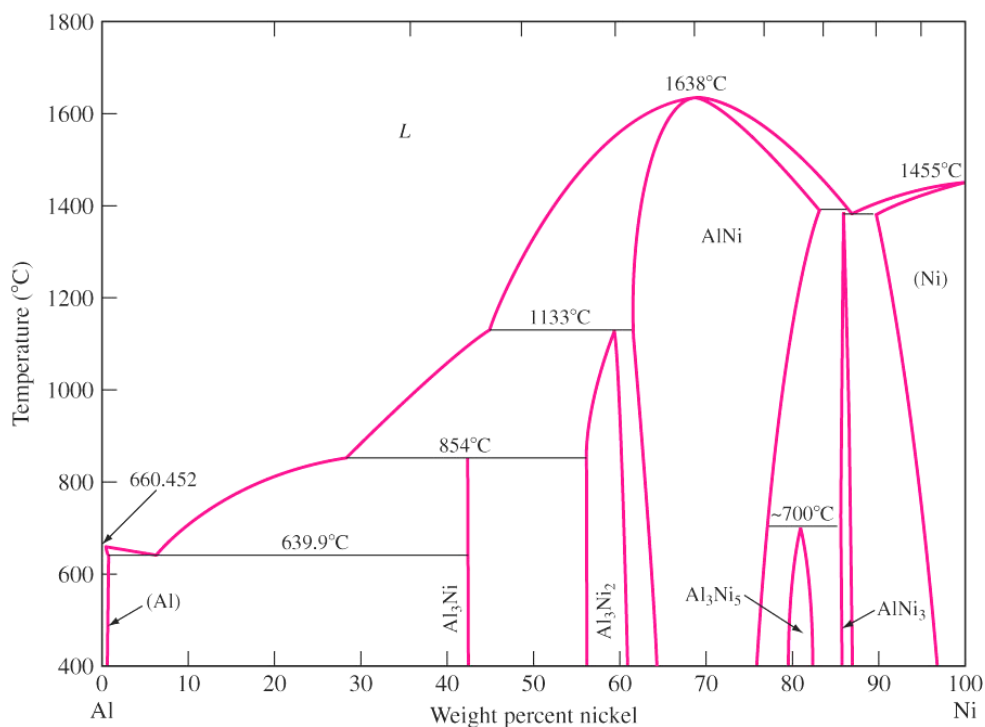
- Write down the invariant reactions, indicating the temperature, the reaction in equation form, the composition of each phase and the name of the reaction.

Solution: eutectic reaction (639°C; 1380°C); peritectoid reaction (700°C); peritectic reaction (854°C; 1133°C, 1400°C)

- Given an Al-20% Ni alloy, indicate at 400°C:

- Composition and mass fraction of the phases.
- Composition and mass fraction of the microconstituents.
- Draw the microstructure that can be expected when cooling in equilibrium conditions. Point out each microconstituent with arrows.

Solution: $C_{Al3Ni}=43\%Cu$; $C_\alpha=1\%Cu$; $X_\alpha=0.55$; $X_{Al3Ni}=0.45$; Microconstituents: eutectic ($X=0.64$) and proeutectic Al_3Ni ($X=0.36$).



10. Considering the Cu-Sn phase diagram:

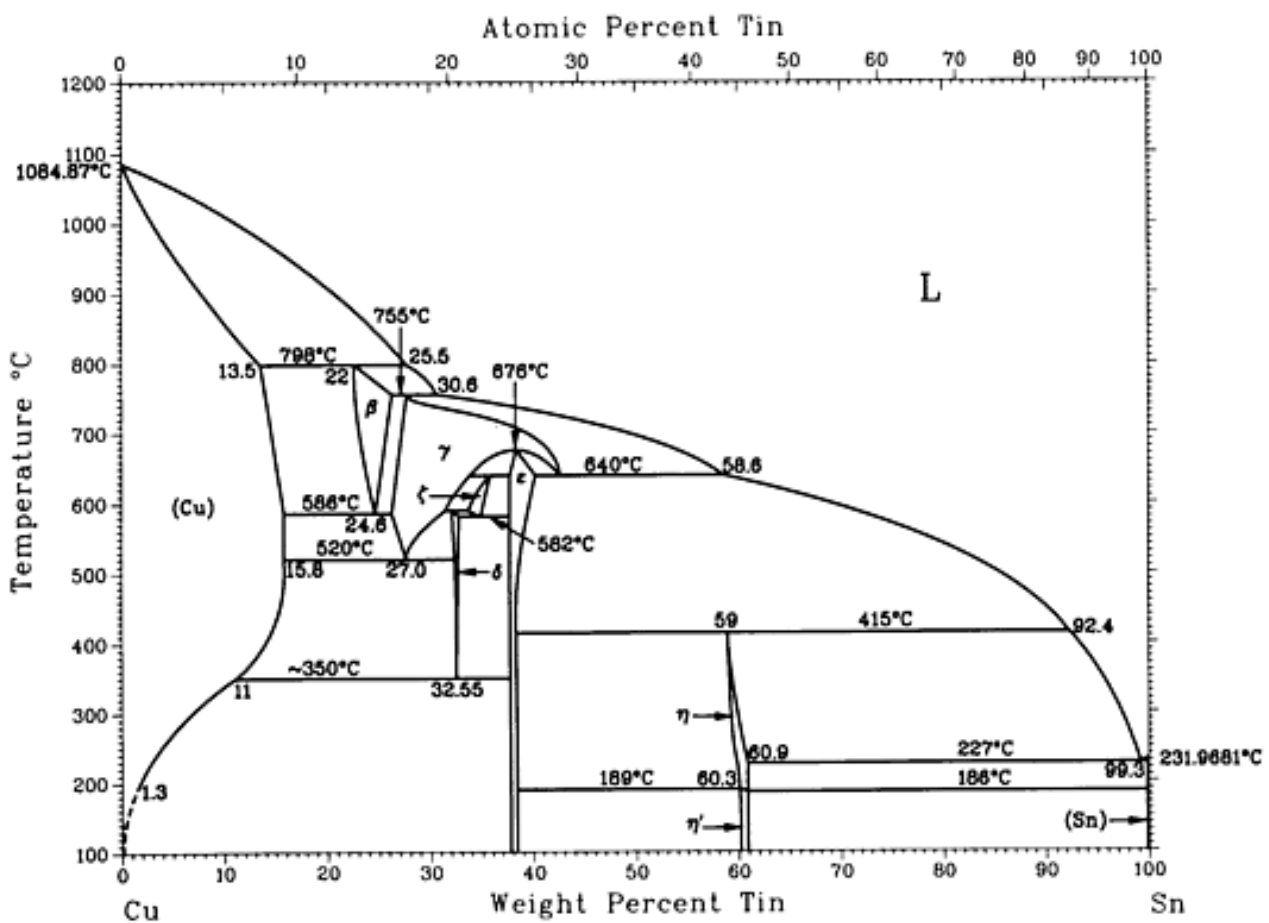
- Indicate the type of transformations, and write the reactions that take place at $T_1 = 798^\circ\text{C}$; $T_2 = 755^\circ\text{C}$; $T_3 = 588^\circ\text{C}$; $T_4 = 415^\circ\text{C}$; $T_5 = 1084.87^\circ\text{C}$. (**peritectic; peritectic, eutectoid; peritectic and solidification of Cu**)
- For a Cu-20%Sn alloy, indicate the phases and microconstituents, their composition and amount at $T_1: 521^\circ\text{C}$ and $T_2: 519^\circ\text{C}$.

Solution:

521°C : $C_{(\text{Cu})} = 15.8\%\text{Sn}$; $C_\gamma = 27\%\text{Sn}$; $X_\gamma = 0.37$; $X_{(\text{Cu})} = 0.63$; $X(\text{eutectoid}) = 0.48$; $X(\text{proeutectoid Cu}) = 0.52$

519°C : $C_{(\text{Cu})} = 15.8\%\text{Sn}$; $C_\delta = 32\%\text{Sn}$; $X_\delta = 0.26$; $X_{(\text{Cu})} = 0.74$; $X(\text{eutectoid}) = 0.37$ and $X(\text{proeutectoid Cu}) = 0.63$

- Determine the percentage of Sn in an alloy having 54% liquid phase at 900°C . Is this alloy in solid state at 700°C ? (**14.5%Sn**)



11. Draw the hypothetical phase diagram of A and B from the following data.

- Melting temperatures: $A = 850^\circ\text{C}$; $B = 350^\circ\text{C}$
- A and B form an stoichiometric compound AB_2 that melts at 1000°C ; neither A nor B are soluble in the compound.
- There is an invariant eutectic point (70% at. of A, 700°C).
- There is an invariant peritectic point (80% at. of B, 500°C).
- The composition of the liquid at 500°C is 95% at. of B.
- The solubility of B in A is 10% at. at 700°C and 2% at. at 25°C .
- The solubility of A in B is 25% at. at 25°C .

Connect the points with straight lines. Indicate all the phase areas and points in the diagram, and write the reactions of the three phases

12. According to the Ni-Mo phase, indicate:

- a. The equations that represent the invariant reactions, specifying the composition of the phases, temperature and type of reaction.

Solution: eutectic reaction (1310°C); peritectoid reaction (910°C); peritectic reaction (1362°C).

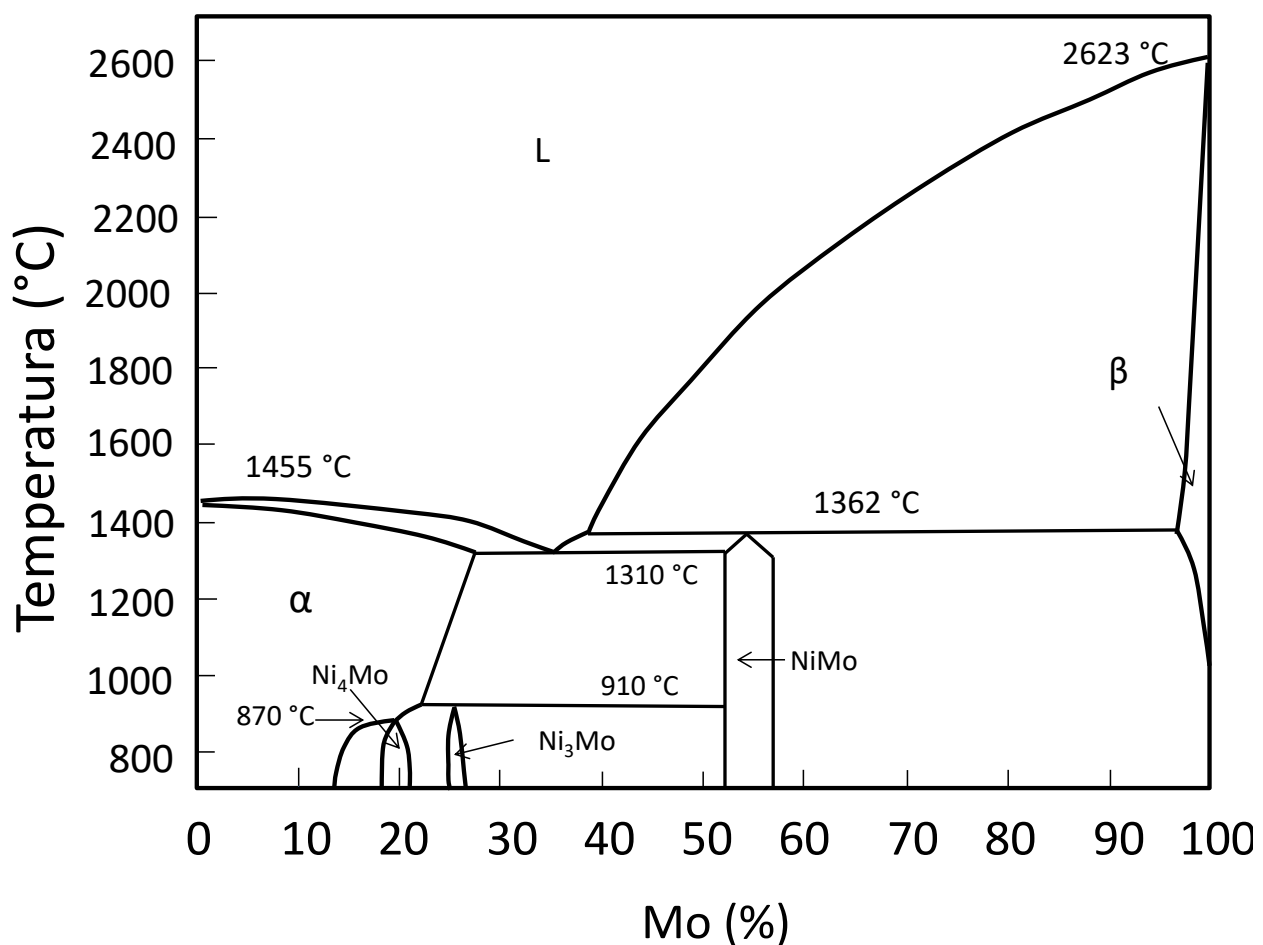
- b. The maximum solubility of Mo in Ni and at what temperature it is produced (**26% Mo**)
 c. The maximum solubility of Ni in Mo and at what temperature it is produced (**4% Ni**)
 d. Melting temperature of Ni and Mo (**1455°C; 2623°C**)

- e. Present phases, their composition and amount for a Ni-40%Mo alloy at 1309°C.

Solution: $C_{NiMo}=52\%Mo$; $C_{\alpha}=27\%Mo$; $X_{\alpha}=0.48$; $X_{NiMo}=0.52$

- f. Microconstituents and their amount for a Ni-40%Mo alloy at 1309°C. Sketch the microstructure

Solution: $X_{NiMoP}=0.25$; $X_E=0.75$



13.- Given the following Ti-Si phase diagram answer the next questions:

- a) Indicate the phases or compounds with incongruent and congruent melting (**solution: congruent melting: Ti_5Si_3 and $TiSi_2$ incongruent melting: Ti_3Si , Ti_5Si_4 and $TiSi$**)

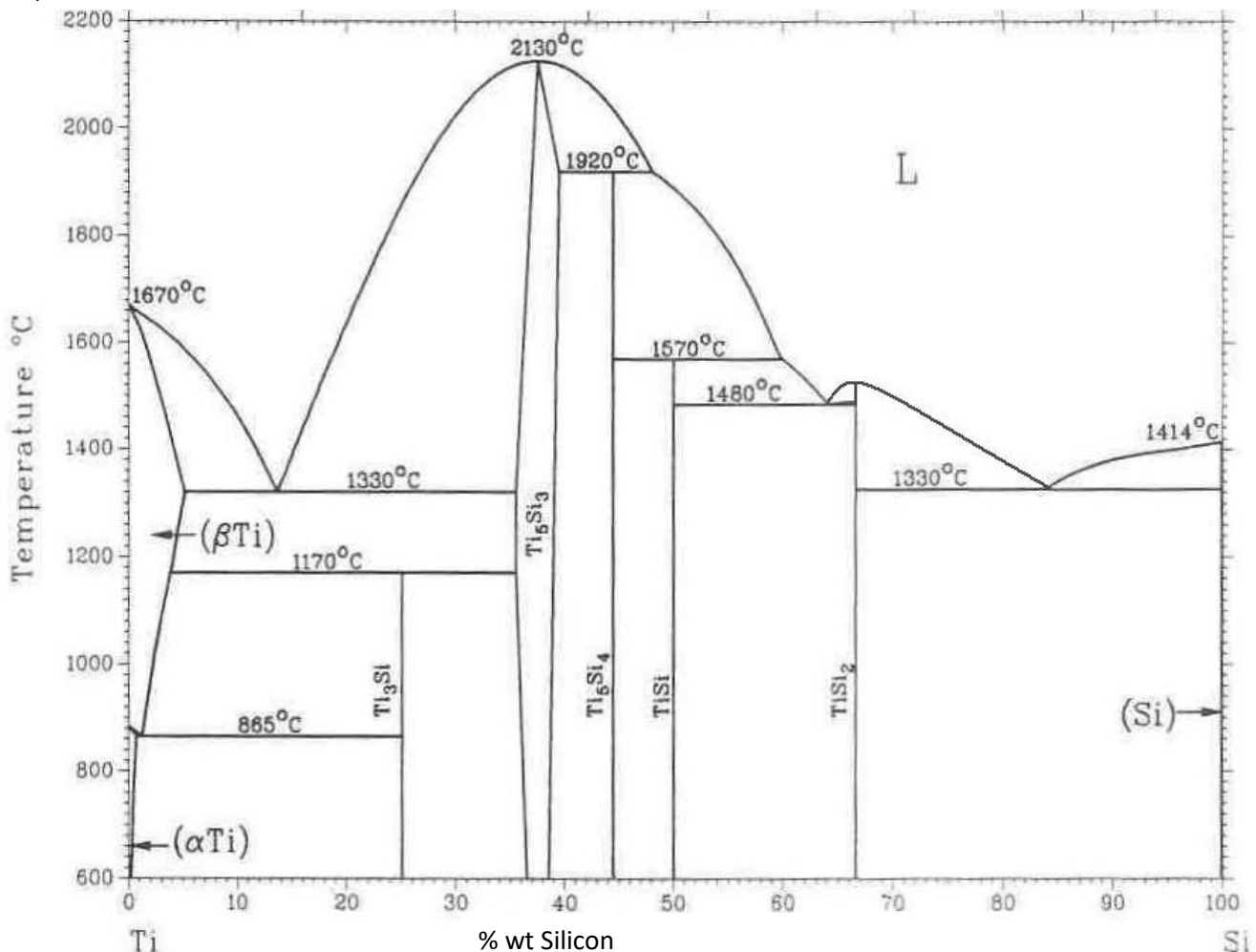
- b) Write down the invariant reactions, specifying their name, temperature, and composition of the phases present. (**solution: Eutectic, $T = 1330^\circ C$; Eutectic, $T = 1330^\circ C$; Eutectic, $T = 1480^\circ C$; Peritectic, $T = 1920^\circ C$; Peritectic, $T = 1570^\circ C$; Peritectoid $T = 1170^\circ C$; Eutectoid, $T = 865^\circ C$**)

- c) For an alloy with 30% Si describe the solidification and cooling sequence to 600 °C. Describe specifically the microstructure that can be expected at 1329 °C, 1169 °C, and 600 °C. (**solution: $T = 1329^\circ C$ $C_{BTi} = 5.5\% Si$**)

and $C_{Ti5Si3} = 35\% \text{ Si}$, $x_{\beta Ti} = 0.17$, $x_{Ti5Si3} = 0.83$; $T = 1169^\circ\text{C}$, $C_{Ti3Si} = 25\% \text{ Si}$ and $C_{Ti5Si3} = 35\% \text{ Si}$, $x_{Ti5Si3} = 0.5$ $x_{Ti3Si} = 0.5$)

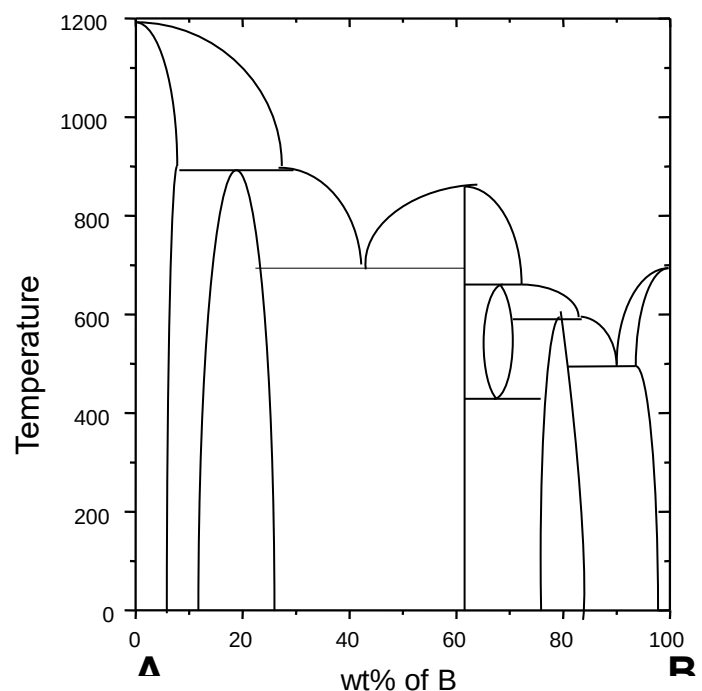
d) For the same alloy of part c) and the temperatures of 1329°C and 1169°C indicate phase composition, phase fractions and quantity of microconstituents present. (solution: $T = 1329^\circ\text{C}$, $x_{\text{eutecti}} = 0.24$, $x_{Ti5Si3 \text{ primary}} = 0.76$; $T = 1169^\circ\text{C}$ phases and microconstituents are the same)

e)



14. In the following graph, a hypothetical phase diagram is represented, identify and explain the following:

- The existing phases in each phase field of the diagram.
- The invariant reactions writing the temperature and the reaction's equation, the composition of each phase inside the reaction and its name.
- The congruent and incongruent fusion phases.



15. Given the phase diagram of the figure, answer the following questions:

- a. Write down the invariant reactions, indicating the name, temperature and composition of the phases involved.

Solution: peritectic (2200°C); peritectic (2190°C); eutectic (1540°C); eutectic (2080°C)

- b. List the congruent and incongruent melting phases or compounds.

Solution: $\text{Ti}_3\text{B}_4 \rightarrow \text{L} + \text{TiB}_2$

- c. For an alloy with 10% B, indicate the composition of the phases, the amount of phases and the amount of microconstituents at 1000°C and 500°C.

Solution:

1000°C: $C_{(\beta\text{Ti})} = 0\% \text{B}$; $C_{\text{TiB}} = 18\% \text{B}$; $X_{(\beta\text{Ti})} = 0.44$; $X_{(\text{TiB})} = 0.56$; eutectic ($x = 50\%$) and primary TiB ($x = 50\%$)

500°C: $C_{(\alpha\text{Ti})} = 0\% \text{B}$; $C_{\text{TiB}} = 18\% \text{B}$; $X_{(\alpha\text{Ti})} = 0.44$; $X_{(\text{TiB})} = 0.56$; eutectic ($x = 50\%$) and primary TiB ($x = 50\%$)

- d. Describe the solidification sequence of an alloy with 18% B at a very slow cooling from the liquid state up to 500°C.

